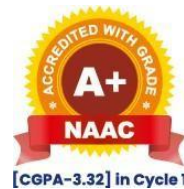




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Project Guidelines:

Common Instructions

In the project documentation the sequence of preamble is

- a) Title Page in specified format**
- b) Institute Certificate**
- c) Internship / Company Certificate**
- d) Guide certificate**
- e) Acknowledgement**
- f) Index with page numbers**

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Font size: 13pts

Line spacing: 1.5 pt

Left Margin: 1.5 inches

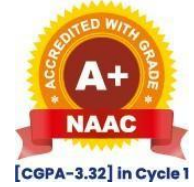
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1. **Application Development Project**

Chapter No		Details
1		Introduction
	1.1	Company Profile
	1.2	Problem Statement
	1.3	Objectives of Proposed System
	1.4	Scope of the Proposed System
2		Analysis and Design
	3.1	Entity Relationship Diagram (ERD)
	3.2	Business Use Case Diagram
	3.3	Table Structure/Data Dictionary
3		Testing
	4.1	Test Cases with results
	4.2	Defect report / Test Log
4		Limitations of Proposed System
5		Proposed Enhancements
6		References
7		User Manual (All input and output screens with consistent data)

2. Cloud Computing

Project Requirement	Details
Cloud Environment Setup	Use AWS, Azure, GCP, or OpenStack
Service Model	Specify IaaS, PaaS, SaaS, or FaaS
Deployment Model	Public, Private, Hybrid, or Multi-Cloud

Index: Cloud Computing

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Company/Institute/Client Profile	
1.2	Abstract	
1.3	Existing System and Need for System	
1.4	Scope of System	
1.5	Objectives	
1.6	Operating Environment (Hardware/Software)	
1.7	Brief Description of Technology Used	
Chapter 2	Technology Used	
2.1	Overview of Study Involved	
2.2	Evaluation of Existing Models	
Chapter 3	Design and Implementation	
3.1	Cloud Service Provider Setup (AWS/Azure/GCP)	
3.2	Performance Metrics (High Availability, Fault Tolerance)	
Chapter 4	Security Implementation	
4.1	Data Encryption, Multi-Factor Authentication, RBAC	
Chapter 5	Deployment, Scalability, and Conclusion	
5.1	Testing and Deployment	
5.2	Interpretation of Results	
5.3	Limitations and Challenges	
5.4	Summary of Objectives and Achievements	
Chapter 6	References	
Chapter 7	Appendices	
Chapter 8	Annexure- Progress Sheet	

3. Cyber Security:

Project Requirement	Details
Threat Analysis	Identify common and advanced threats (e.g., APTs, ransomware)
Security Controls	Implement network security, IAM, and data protection
Research Focus	Choose AI, IoT, Cloud, or Human Factors in Cybersecurity

Index: Cyber security

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Company Profile	
1.2	Abstract	
1.3	Cybersecurity Fundamentals 1.3.1 Cybersecurity Fundamentals 1.3.2 Cybersecurity Landscape 1.3.3 Cybersecurity Challenges in Different Sectors	
1.4	Cybersecurity Landscape	
1.5	Cybersecurity Challenges in Different Sectors	
Chapter 2	Literature Review	
2.1	Overview of Existing Research	
2.2	Identification of Research Gaps	
2.3	Relevance to Research Focus	
Chapter 3	Threat Landscape Analysis	
3.1	Common Cyber Threats	
3.2	Advanced Persistent Threats (APTs)	
3.3	Threat Actor Profiling	
3.4	Vulnerability Assessment	
Chapter 4	Security Control and Technologies	
4.1	Network Security	
4.2	Endpoint Security	
4.3	Identity and Access Management (IAM)	
4.4	Data Protection	
4.5	Security Information and Event Management (SIEM)	

4. AI/ML/DL/Data Science

Project Areas

Students may choose projects in, but not limited to, the following areas: Machine Learning (ML) and Deep Learning (DL), Natural Language Processing (NLP), Computer Vision, Predictive Analytics, Big Data Analytics, Reinforcement Learning, AI-driven Automation, Data Visualization, AI Ethics and Responsible AI

Project Requirement	Details
Data Collection	Use real-world datasets or APIs
Model Development	Implement ML algorithms (e.g., regression, classification)
Evaluation	Use metrics like accuracy, precision, recall, and F1-score

Index: AI/ML/DL/Data Science

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	Dataset Description	
Chapter 2	Literature Review	
2.1	Existing Research	
2.2	Research Gaps	
Chapter 3	Methodology	
3.1	Data Preprocessing	
3.2	Model Selection	
3.3	Implementation	
Chapter 4	Results and Discussion	
4.1	Model Performance	
4.2	Visualization	
4.3	Insights	
Chapter 5	Conclusion	
5.1	Summary	
5.2	Future Work	
Chapter 6	References	
Chapter 7	Appendices	
Chapter 8	Annexure- Progress Sheet	

5. Guidelines for Entrepreneurship Project

Project Areas

Students aiming to foster an entrepreneurial ecosystem to become an entrepreneur with the essential skills, knowledge, and mindset required to identify market opportunities and successfully launch new ventures. Following are the guidelines to be followed by aspiring students for the Entrepreneurship development Projects:

Index: Entrepreneurship Development Project

Chapter	Content	Page Number
Chapter 1	Introduction of the startup	
1.1	Introduction of the organization	
1.2	Vision and mission statement	
1.3	Business objectives	
1.4	Nature of the business	
1.5	Designation, Role and Responsibility in the organization	
1.6	Udyam Certificate	
1.7	Validity Certificate	
1.8	Shopact License	
1.9	Udyam Certificate	
1.10	GST Certificate (Optional)	
Chapter 2	Business Idea / Startup Concept	
2.1	Brief introduction of the business idea	
2.2	Problem being solved	
2.3	Proposed solution	
2.4	Target market	
2.5	Expected impact	
2.6	Problem Statements	
2.7	Description of the product/service	
2.8	Key features	
2.9	Unique value proposition (USP)	
Chapter 3	Market Analysis	
3.1	Target Customers	
3.2	Market Size and Opportunities	
3.3	Competitor Analysis	
3.4	Market trends	
3.5	Promotion Methods	
3.6	Digital Marketing Plan	
Chapter 4	Business Model	
4.1	Value Proposition	
4.2	Revenue Model	
4.3	Pricing strategy	
Chapter 5	Technical Requirements	

5.1	Technology Stack	
5.2	Development process (Software / Tools)	
5.3	System Architecture (if applicable)	
5.4	System architecture (if it is a software startup)	
Chapter 6	Financial Plan	
6.1	Estimated Investment	
6.2	Cost Analysis	
6.3	Profit Forecast	
Chapter 7	Risk Analysis	
7.1	Potential Risks	
7.2	Risk Mitigation Strategies	
Chapter 8	Implementation Plan	
8.1	Development Phases	
8.2	Timeline	
Annexure 1	Expected Outcomes	
	User Manual (All input and output screens with consistent data)	
Annexure 2	Conclusion	
Annexure 3	Future Scope	
Annexure 4	References / Bibliography	
Annexure 5	Appendix	

6. AI Agenting

Project Areas

Students having project in may choose projects in, but not limited to, the following areas: Machine Learning (ML) and Deep Learning (DL), AI and creating its agents. Students undertaking projects in AI Agenting are required to follow the guidelines given below for effective planning, development, and implementation of their work.

Project Requirement	Details
Data Collection	Use real-world datasets or APIs
Model Development	Implement ML algorithms, AI agents (e.g., regression, classification)
Evaluation	Use metrics like accuracy, precision, recall, and F1-score, AI agents

Index: AI/ML/DL/Data Science/AI agents

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	Dataset Description	
1.4	Agentic System Overview (Describe agent-based architecture, Reasoning loop, autonomy and tool usage)	
Chapter 2	Literature Review	
2.1	Existing Research	
2.2	Research Gaps	
2.3	Agent based AI Systems (Architecture/Framework used for Autonomus agents and reasoning agents/systems)	
2.4	LLM Frameworks and Tools (Specific Technology used for agenting as LangChain, CrewAI, AutoGen, RAG etc)	
Chapter 3	Methodology	
3.1	Data Preprocessing	
3.2	Model Selection	
3.3	Agent architecture (Describe reasoning loop, planner and executor components)	
3.4	Prompt Engineering Strategy (Explain system prompts, role prompting and task instructions)	
Chapter 4	Implementation	
4.1	System Implementation	
4.2	Results (System Outputs)	
4.3	Tool Integration (Integration with APIs, databases, embeddings, retrieval pipelines)	
4.4	Knowledge Retrieval System (RAG) Explain vector databases, embeddings and database	

	pipelines)	
Chapter 5	Evaluation	
5.1	Model Performance (Accuracy, Precision, F1 score, Recall)	
5.2	Agent Task Success Rate (Measure how often the agent completes the task correctly)	
5.3	Response Quality Analysis (Evaluate reasoning quality, relevance, hallucinations)	
5.4	Visualization (Charts, Graphs, Dashboards)	
5.5	Insights (Observation from experiments)	
Chapter 6	Conclusion	
6.1	Summary Conclusion	
6.2	Future Work	
6.3	Limitations of the agents	
Chapter 7	References	
Chapter 8	Appendices (Include system prompts used in the agent, sample agent interactions-user queries/responses)	
Chapter 9	Annexure	

7. UIUX

Project Areas

The UI/UX Project Documentation Index outlines a structured approach for documenting the design and development of user-centred digital solutions. It guides students in presenting research, wireframes, prototypes, and usability considerations aligned with industry practices in User Experience Design and User Interface Design.

Project Requirement	Details
Data Collection	Analytics to understand user behavior and requirements in User Experience Design
Model Development	Develop user flows, wireframes, and high-fidelity prototypes applying principles of User Interface Design to create intuitive and user-friendly interfaces
Evaluation	Perform usability testing and evaluate using metrics like user satisfaction, task completion rate, and efficiency in User Experience Design

Index: UIUX

Chapter	Content	Page Number
Chapter 1	Introduction	
1.1	Problem Statement	
1.2	Objectives	
1.3	UX Methodology	
Chapter 2	Literature Review	
2.1	Existing System Analysis	
2.2	User Research	
2.3	Stakeholder Insights	
2.4	Competitive Analysis	
2.5	Key Insights	
Chapter 3	Problem Definition	
3.1	Target Users	
3.2	Personas	
3.3	Empathy Mapping	
3.4	User Journey	
3.5	UX Goals	
Chapter 4	Product Strategy and Structure	
4.1	Proposed Solutions	
4.2	Functional Requirements	
4.3	Non-Functional Requirements	
4.4	Information Architecture	
4.5	Sitemap and Navigation	
4.6	User / Task Flows	
Chapter 5	Interaction and Visual Designs	

5.1	Wireframes	
5.2	Prototypes	
5.3	UI Guidelines	
5.4	Visual Designs	
5.5	Design Systems	
Chapter 6	Usability Testing & Iteration	
6.1	Testing Objectives	
6.2	Testing Method	
6.3	Test Scenarios	
6.4	Findings	
6.5	Iterations (Improvements made based on user feedback.)	
Chapter 7	Implementation and Impact	
7.1	Tools and Technologies	
7.2	Design Handoff	
7.3	UX Metrics	
7.4	Results and Improvements	
7.5	Future Scope	
Chapter 8	References	
Chapter 9	Appendices	
Chapter 10	Annexure	

PROJECT REPORT

ON

NAME OF THE SYSTEM

FOR

NAME OF THE COMPANY

BY

NAME OF STUDENT



UNIVERSITY OF PUNE

MASTER IN COMPUTER MANAGEMENT



**Maharashtra Education Society's
INSTITUTE OF MANGEMENT AND CAREER COURSES
(IMCC), PUNE-411038**